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REVIEW

Molecular Genetics and Probiotic Mechanisms of *Saccharomyces cerevisiae* var. *boulardii*

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Abstract
Saccharomyces cerevisiae var. *boulardii* (*Sb*) is a *S. cerevisiae* (*Sc*) strain that has been widely used in the treatment of gastrointestinal diseases due to its unique probiotic properties. The key genomic differences that distinguish *Sb* from *Sc* include the tetrasomy of chromosome XII, the absence of intact transposon-yeast (Ty) elements, and variations in the copy number of specific genes. These genomic variations may contribute to enhanced thermotolerance, increased acid resistance, and elevated acetate production, collectively supporting its probiotic functions. The probiotic mechanisms of *Sb* are mediated through luminal actions, mucosal actions, and trophic effects. Its luminal activity involves neutralizing pathogen toxins via the secretion of proteins and inhibiting pathogen growth through the production of short-chain fatty acids (SCFAs). Additionally, *Sb* modulates gut microbiota composition by fostering symbiotic relationships, thereby increasing the abundance of beneficial microbes and SCFA levels to promote gut health. The mucosal action of *Sb* promotes anti-inflammatory responses by regulating the nuclear factor kappa B (NF-κB) and mitogen-activated protein kinase (MAPK) pathways. Meanwhile, its trophic effects, driven by polyamine production, enhance the function of intestinal epithelial cells. Recent findings further suggest that *Sb* may serve as a potential adjuvant therapy for brain disorders by modulating the gut–brain axis (GBA) to attenuate neuroinflammation. With continued multidisciplinary research, *Sb* is well-positioned to advance the biotherapeutic landscape. This review aims to synthesize recent advances in the genetics and probiotic mechanisms of *Sb*, with particular emphasis on its modulatory effects on the GBA.

Keywords *Saccharomyces cerevisiae* var. *boulardii* · *Saccharomyces boulardii* · Probiotics · Yeast · Gastrointestinal disorders · Gut-brain axis

Introduction

The species designation of *Saccharomyces boulardii* was coined after a French agricultural engineer and microbiologist Henri Boulard, who isolated the yeast in 1923 from lychee and mangosteen skins after observing that local populations in Indochina consumed fruit-based teas to treat cholera. The French pharmaceutical company Biocodex acquired the intellectual property rights of this *S. boulardii* strain in 1947 and registered it as *S. boulardii* CNCM I-745 in 1953 at the Pasteur Institute. Currently, *S. boulardii* is the only probiotic yeast for the treatment of antibiotic-associated diarrhea, and pediatric acute gastroenteritis, and has clinically proven to be effective for acute and chronic digestive tract disorders, such as *Clostridioides difficile*–induced diarrhea, traveler's diarrhea, irritable bowel syndrome (IBS), and Crohn's disease [1]. *S. boulardii* was initially considered a distinct species from *S. cerevisiae* due to its unique probiotic properties against *C. difficile* infection, inability to metabolize galactose, and distinct mitochondrial DNA fingerprint compared to *S. cerevisiae* [2, 3]. However, galactose metabolism is variable among *S. cerevisiae* strains [4], indicating that biochemical traits such as galactose utilization should not be solely relied upon for the species identification of *S. boulardii* [3].

McCullough et al. [5] argued that DNA-based analyses are more appropriate for the species identification of *S. boulardii*. Two independent studies employing restriction fragment length polymorphism (RFLP) analysis of Intergenic

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Transcribed Spacer (ITS) regions and/or ribosomal DNA failed to genotypically distinguish *S. boulardii* from *S. cerevisiae* [5, 6]. Furthermore, sequence analysis of the D1/D2 domain of 26S rDNA, ITS1-5.8S rDNA-ITS2 regions, and the cytochrome c oxidase subunit 2 (*COX2*) gene across eight *S. boulardii* strains revealed 96.6–100% similarity with *S. cerevisiae* strains [7]. Additional evidence includes the ability of *S. boulardii* to produce fertile hybrids upon crossing with *S. cerevisiae* [8]. Collectively, these findings support the conclusion that *S. boulardii* is not a distinct species but rather a strain of *S. cerevisiae*, with the correct nomenclature being *Saccharomyces cerevisiae* var. *boulardii*. For conciseness, *S. cerevisiae* var. *boulardii* will be abbreviated as *Sb* and *S. cerevisiae* as *Sc* throughout this review. This review aims to discuss the latest molecular insights into *Sb* and its probiotic mechanisms, with a particular focus on its modulation of the gut-brain axis.

Genomic Differences Between *Sb* and Other *Sc* Strains

The first draft genome of *Sb* (Econorm from Dr. Reddy's Laboratory [EDRL], also known as CNCM I-745) was reported by Khatri et al. [9] in 2013. Subsequently, a draft genome of *Sb* ATCC MYA-796 was published [10]. Genomes of four additional *Sb* strains (Biocodex, Unique 28, Kirkman, and Unisankyo) were also sequenced, revealing a high degree of genome similarity to *Sc*, with over 99% of shared Average Nucleotide Identity (ANI) [11]. To date, the genomes of more than 20 *Sb* strains have been sequenced (Table 1), facilitating research into their unique phenotypic traits [9, 11–14], infection mechanisms of food-related yeasts [15], and the development of genetic engineering approaches for *Sb* [16]. Moreover, comparative genome analysis of *Sb* aids in tracing the genomic origins of clinical *Sc* isolates, offering insights into the pathogenic potential of food-related yeasts as opportunistic pathogens [15]. Despite the high genomic similarity between *Sb* and *Sc*, subtle genomic differences contribute to their phenotypic divergence, as illustrated in Fig. 1. The following section discusses these genomic and phenotypic distinctions to underscore the uniqueness of *Sb* within the broader context of *Sc* diversity.

Analytical Methods for Distinguishing *Sb* and *Sc* Strains

Several molecular methods have been developed to distinguish *Sb* from other *Sc* strains. Microsatellite typing, which amplifies and detects the polymorphic dinucleotide and trinucleotide repeats, successfully differentiated *Sb* based on a unique trinucleotide polymorphism—nine CAG repeats within the *YLR177w* locus [17]. Additionally, Posteraro et al. [18] applied the Southern blotting-based Ty917 hybridization method, achieving higher distinguishing power for *Sb* identification. Furthermore, combined polymerase chain reaction (PCR) amplification of interdeltas, ITS regions, and microsatellites enables rapid multiplex strain identification of *Sb* [19]. More recently, a quantitative PCR high-resolution melting (qPCR-HRM) assay targeting the putative aryl-alcohol dehydrogenase (*AAD15*) gene has been developed to differentiate *Sb* from closely related *Sc* strains [20].

Aneuploidy

Comparative genomic hybridization (CGH) analysis initially suggested that *Sb* exhibits trisomy of chromosome IX relative to *Sc* [8]. However, subsequent PacBio long-read sequencing of the *Sb* genome contradicted this finding [11], revealing instead a double read coverage of chromosome XII, indicating tetrasomy of chromosome XII in *Sb* [11].

Lack of Certain Intact Ty Elements

CGH analysis and PCR amplification with the primers targeting long terminal repeats (LTRs) of Ty revealed that *Sb* has lost most intact Ty1 or Ty2 elements [21]. Subsequent PacBio long-read genome sequencing confirmed the absence of intact Ty1, Ty3, and Ty4 elements in seven *Sb* strains, although LTRs of Ty1/2 (δ), Ty3 (σ), Ty4 (τ), and Ty5 (ω) remain present [11]. The loss of these Ty elements may be linked to the higher physiological temperature (37°C) of *Sb*, which inhibits the Ty1 protease (PR) responsible for cleaving the Gag-Pol polyprotein and the reverse transcriptase (RT) required for transcribing Ty1 RNA into cDNA, ultimately preventing the transposition of Ty elements [22, 23]. Additionally, the diploid nature of *Sb* may contribute to the absence of certain intact Ty elements, as Ty1-RNA levels in *MATa/α* diploids are found to be 20-fold lower than in *Sc* haploids [24]. Therefore, the transposition rate of Ty elements in *Sb* is inherently low due to both the elevated growth temperature and the repressed transcription associated with its *MATa/α* diploid configuration.

Furthermore, LTR-LTR recombination in *Sb* might explain the absence of intact Ty elements. Such recombination events have been reported in *Sc* and Ty1-less *S. paradoxus*, where two nearly identical LTRs flanking a Ty element mediate crossover, resulting in excision of the flanking sequence and the retention of a solo LTR [25, 26]. Due to the higher rate of Ty element loss through LTR-LTR recombination and the lower transposition rate of Ty element, *Sb* might have experienced a prolonged period of element loss, eventually leading to the complete loss of intact Ty elements. This hypothesis explains the detection of LTRs in the *Sb* genome, while intact Ty elements remain undetected.

Table 1 Summary of genomics and transcriptomics sequencing on *Sb*

Strain	Omics	Sequencing platform	Sequence Read Archive (SRA)	*WGS accession	BioProject/Gene Expression Omnibus (GEO)	References
EDRL (CNCM I-745)	Genome	LS454 (454 GS FLX Titanium)	SRX424541	ATCS00000000	PRJNA207020	[9]
		Illumina HiSeq	-	ATCS02000000		[11]
ATCC MYA-796	Genome	Illumina HiSeq	-	JRHY00000000	PRJNA260553	[10]
Sb-17	Genome	Illumina MiSeq	-	JPJH00000000	PRJNA255142	Unpublished by [10]
ATCC MYA-797 (Ultra Levure)	Genome	Illumina HiSeq	-	LRVB00000000	PRJNA308002	[12]
Biocodex	Genome	PacBio P6C4	-	LIIL01000000	PRJNA259301	[11]
	Transcriptome	Axon 4000B scanner	-	-	PRJNA185195/ GSE43271	[110]
		Illumina NextSeq	-	-	PRJNA648307/ GSE155032	[43]
Unique28	Genome	PacBio P6C4	-	LIOO01000000	PRJNA285730	[11]
Kirkman	Genome	Illumina HiSeq	-	LOMX01000000	PRJNA259300	[11]
Unisankyo	Genome	Illumina HiSeq	-	LNQF01000000	PRJNA274654	[11]
7103	Genome	Illumina HiSeq	SRX5874555	-	PRJNA542592	[13]
Sb.P	Genome		SRX5874554			
SAN	Genome		SRX5874553			
ENT	Genome		SRX5874552			
FLO	Genome		SRX5874551			
Inferior pool—acetic acid/boulardii	Genome		SRX5874550			
Superior pool—acetic acid/boulardii	Genome		SRX5874549			
Hybrid SBERH6	Genome		SRX5874548			
7136	Genome		SRX5874547			
259	Genome		SRX5874546			
UL	Genome		SRX5874545			
Sb.L	Genome		SRX5874544			
7135	Genome		SRX5874543			
LSB	Genome		SRX5874542			
Sb.A	Genome		SRX5874541			
KCTC 13826BP	Genome	PacBio	SRX17913227	JAPJTW000000000	PRJNA891108	[40]
AQ2584 (Ultralevura, haploid)	Genome	Illumina NextSeq	SRX18673526	-	PRJNA910339	[15]
CLPY01	Genome	Illumina HiSeq	-	JXBM00000000	PRJNA271636	[16]
Ent ancestral	Genome	Illumina HiSeq	SRX18783832	-	PRJNA911196	[14]
Ent ev16			SRX18644802			
Ent ev17			SRX18765449			

*Whole genome sequencing

Leveraging the LTRs, Liu et al. [27] successfully applied the EasyCloneMulti system [28], originally developed for *Sc* Ty integration to *Sb* for the multicopy integration of enhanced green fluorescent protein (EGFP) into the LTR sites of Ty1.1 (52 sites), Ty1.2 (37 sites), Ty2 (16 sites), and Ty3 (35 sites). Remarkably, integration into Ty2 LTRs gives the highest transformation efficiency and produces strains with significantly higher expression of enhanced green fluorescence protein, despite the lowest number of LTR sites in the *Sb* genome.

Gene Copy Number Variations

According to CGH analysis, variations in gene copy numbers between *Sb* and *Sc* predominantly occur within the subtelomeric regions of the chromosomes [8]. Notably, *Sb* exhibited

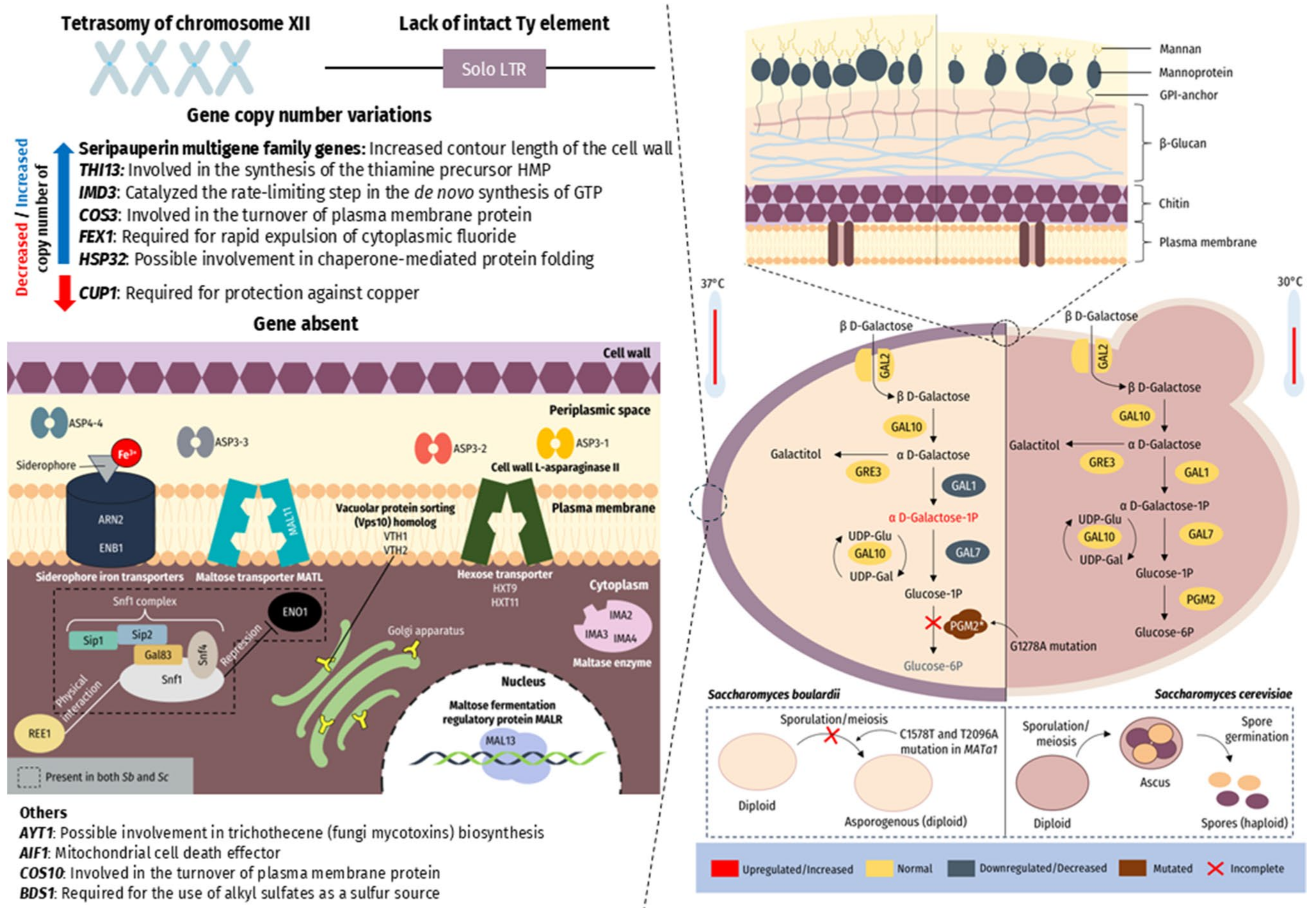

Fig. 1 Genomic and phenotypic differences between *Sb* and *Sc*. *Sb* is a diploid strain characterized by tetrasomy of chromosome XII, whereas *Sc* exists in both haploid and diploid forms. Unlike *Sc*, *Sb* lacks intact Ty transposable elements, retaining only solo long terminal repeats (LTRs). *Sb* displays gene copy number variations, including increases in *THI13*, *IMD3*, *COS3*, *FEX1*, and *HSP32*, and a decrease in *CUP1*. Additional variations are observed in genes associated with maltose utilization (*MAL11*, *MAL13*, *IMA2*, *IMA3*, *IMA4*), hexose metabolism (*HXT9*, *HXT11*, *REE1*), asparagine catabolism (*ASP3-1*, *ASP3-2*, *ASP3-3*, *ASP3-4*), siderophore iron transport (*ARN2*, *ENB1*), vacuolar sorting (*VTH1*, *VTH2*), and others (*AYT1*, *AIF1*, *COS10*, *BDS1*). *Sb*'s thicker mannoprotein layer of the cell wall—likely attributed to the expanded seripauperin multigene family—enhances acetate resistance. A G1278A point mutation in *PGM2* reduces galactose metabolism but supports improved growth at 37°C. Finally, *Sb* is asporogenous, likely due to two-point mutations (C1578T and T2096A) in *MATa1*, resulting in incomplete meiosis and spore formation.

2-fold reduction in the copy number of *CUP1* (*YHR053C*), a gene encoding a metallothionein that binds copper, which contributes to the copper sensitivity of *Sb* (unable to grow in the presence of 0.5 mM copper ions) [8]. PacBio long-read sequencing identified 144 genes absent in *Sb*, including 85 dubious open reading frames (ORFs), 32 uncharacterized genes, and 27 functional genes [11]. For instance, genes responsible for maltose utilization (*MAL11* and *MAL13*), palatinose utilization (*IMA2*, *IMA3*, and *IMA4*), and hexose transport (*HXT9* and *HXT11*) are missing in *Sb* [11]. Additionally, *Sb* lacks genes involved in asparagine catabolism (*ASP3*-1, *ASP3*-2, *ASP3*-3, and *ASP4*-4), putative membrane glycoproteins (*VTH1* and *VTH2*), siderophore iron transporters (*ARN2* and *ENB1*), regulator of enolase 1 (*REE1*) involved in the galactose metabolic pathway via Gal83p [29], trichothecene 3-O-acetyltransferase (*AYT1*), apoptosis-inducing factor 1 (*AIF1*), endosomal endocytosis protein (*COS10*), and bacterially derived sulfatase (*BDS1*) [11].

The same study also reported increased copy numbers of specific genes in *Sb* compared to *Sc* strains, which may contribute to *Sb*'s enhanced stress tolerance [11]. PacBio genome sequencing reveals a 5- to 6-fold increase in the copy number of *THI13*, a gene essential for the biosynthesis of the thiamine precursor hydroxymethylpyrimidine (HMP) [30]. A 2- to 4-fold increase in the gene copy number of *IMD3*, which is involved in guanosine-5′-triphosphate (GTP) biosynthesis and mycophenolate resistance, is also detected in *Sb* [31]. Thiamine, derived from HMP, serves as a critical enzyme cofactor and plays a key role in cellular energy metabolism, while GTP is a building block for

RNA synthesis, an energy source for specific biosynthetic pathways, and an activator of G-protein-coupled receptor (GPCR) signaling [32]. Additionally, the copy number of *COS3*, a gene that enhances salinity resistance when overexpressed [33], is found to be 4- to 5-fold higher in *Sb*. The genes *FEX1* (encoding a fluoride exporter) [34] and *HSP32* (encoding a heat shock chaperone protein) exhibit a 3-fold increase in gene copy number, contributing to fluoride resistance and heat stress tolerance, respectively.

Phenotypic Differences Between *Sb* and *Sc* Strains

Sporulation Deficiency in *Sb*

Sporulation is a developmental process in which a *MATa/α* diploid yeast cell undergoes meiosis to form four haploid spores in response to unfavorable conditions, such as nutrient deprivation. These spores, protected by four layered-cell walls, exhibit high resistance to environmental stress and remain dormant until conditions become favorable for germination and vegetative growth. The *MATa/α* diploidy of *Sb* has been confirmed via mating-type PCR amplification of both mating alleles, *MATa* and *MATα* [21]. Despite its *MATa/α* diploidy, *Sb* is asporogenous [5] because spore meiosis cannot be completed (Fig. 1) [8]. Early hypothesis suggests that the sporulation deficiency might be attributed to the sequence divergences in genes such as *MND2* (encoding a subunit of anaphase-promoting complex/cyclosome), *CDC16* (encoding a subunit of the anaphase-promoting complex), and *DMC1* (encoding a meiosis-specific protein involved in double-strand break repair and homologous chromosomes pairing) [8]. However, a comparative genome study of five *Sb* strains found greater than 99% similarity of these genes, as well as in other meiotic and mitotic genes that could potentially hinder the sporulation pathway in *Sb* compared to *Sc* [11].

Khatri et al. [11] proposed that mutations in the *MATa* locus of *Sb* (eight substitutions and seven insertions) are responsible for the asporogenous phenotype, similar to the non-sporulating brewer's yeast strain with polymorphisms in the *MATα* gene [35]. The *MATa* locus encodes the ORFs of MATa1 and MATa2 proteins. Specifically, MATa1 interacts with MATα2 to form the MATa1-MATα2 complex, which represses the expression of mating-type-specific genes [36]. The two mutations (C1587T and T2096A) in the ORF of *MATa1* may disrupt this repressor complex, leading to the retention of haploid-specific gene expression in diploid *MATa/α Sb*, including *GPA1*, *STE4*, *STE12*, *STE18*, and *RME1* [11]. Notably, *RME1* encodes a transcription factor that represses the master activator of meiosis, *IME1* [37–39]. Consequently, the aberrant expression of *RME1* in *Sb* may prevent the completion of meiosis during sporulation.

The asporogenous nature of *Sb* hinders the conventional yeast sporulation-based analysis and complicates genetic studies. To address this limitation, Offei et al. [13] developed *Sb* strains with homozygous mating types (*MATa/a* or *MATα/α*). This was achieved by first generating a heterothallic (unable to switch mating type) strain through homozygous deletion of the homothallic switching endonuclease (*HO*) genes. Subsequently, a plasmid containing a single copy of *HO* under the control of *GAL1* promoter (P_{GAL1}) was introduced into the heterothallic *Sb* strain. Galactose induction of *HO* expression generated stable homozygous mating-type strains. Crossing these homozygous *Sb* strains with *Sc* strains of complementary mating types restored sporulation competence in *Sb*, allowing subsequent crossing of strains for elucidation of *Sb*-specific traits.

Carbon Metabolism Differences

Although *Sb* was initially believed to be incapable of metabolizing galactose, McCullough et al. [5] identified three commercial *Sb* strains, namely Ultra-Levure's Sb 48 and Sb 49 from France and Codex (Sb It) from Italy, capable of galactose metabolism. Similarly, Rajkowska and Kunicka-Styczyńska [3] reported galactose utilization in *Sb* strain CNCM I-745, which was corroborated by a genomics study by Khatri et al. [9], who demonstrated that the strain harbors all the required galactose metabolism genes. Chelliah et al. [40] also reported galactose metabolism in an isolated *Sb* strain from fermented idli (KCTC 13826BP). A recent study found that *Sb* strain ATCC MYA-796 can metabolize galactose, although at a much slower rate than *Sc* BY4742 [41]. Notably, *Sb* consumes less galactose under anaerobic conditions or at higher temperatures (37°C) [41].

Higher galactose concentrations (≥ 2 g/L) are toxic to *Sb*, resembling toxicity observed in *Sc* due to galactose-1-phosphate (Gal-1-P) accumulation from lithium-ion inhibition of phosphoglucomutase (PGM2) [42]. A nonsense mutation (G1278A) in the *PGM2* gene resulted in a truncated PGM2 protein (from 569 to 426 amino acids), which represses the upstream galactose utilization genes *GAL1* and *GAL7* [43]. Reversing this nonsense mutation produces an *Sb* strain that can metabolize galactose more efficiently, but at the cost of losing its growth advantage at 37°C. The mechanism by which the G1278A mutation enhances the growth rate of *Sb* at 37°C remains unclear. Contrasting to galactose, *Sb* is unable to metabolize palatinose [44] due to the absence of palatinose utilization genes [11].

Probiotic Traits and Mechanisms of *Sb*

The first draft genome of *Sb* [9] revealed several candidate genes that may contribute to probiotic properties of *Sb* such as 54 kDa serine protease [45, 46], 120 kDa kinase/transporter involved in the cAMP pathway [47], and 63 kDa phosphatase (*PHO8*) [48]. However, these genes are not unique to *Sb*, as their homologs are also found in *Sc*. *Sb* cannot colonize the host gastrointestinal tract as it is cleared out of the system within 72 h [1]. Nevertheless, *Sb* remains the only FDA-approved probiotic yeast to date, partly owing to its ability to survive and persist in the gastrointestinal (GI) tract. The following sections will explore the specific traits that contribute to its probiotic properties.

Acetate Tolerance and Production

Cell Wall Adaptations

The high acid resistance of *Sb* allows its survival in the harsh GI environment and facilitates its transition further into the GI tract to serve as a probiotic. Unlike *Sc*, *Sb* thrives in the simulated gastric environment [49], on yeast extract peptone dextrose agar of pH 2 [8], or in weak acidic (0.5% acetate) conditions [50]. Gene ontology analysis of the *Sb* ATCC MYA-797 genome unveiled a significant amount of indels or amino acid substitutions in cell wall organization and assembly compared to *Sc*, which may attribute to the thicker cell wall for increased acid resistance [12]. Notably, *Sb* has 18 copies of seripauperin multigene family genes, which are positively correlated with the contour length of the cell wall [51] —a measure of the maximum force required to stretch and extend the cell wall to its breaking point— as compared to seven copies in *Sc* S288C [11]. Furthermore, yeast surface analysis using scanning electron microscopy observed a relatively more intact surface of *Sb* with increased budding compared to *Sc* under 0.5% acetate [50]. This observation suggests that a thicker mannoprotein layer of the cell wall (Fig. 1) protects *Sb* against weak acids [12].

Genetic Basis of Enhanced Acetate Production

A comparative transcriptomics study revealed that *Sb* perceives simulated stomach conditions as less stressful than *Sc*. Specifically, the *Sc* transcriptome displayed a general stress response pattern, with upregulation of carbohydrate/energy metabolism and stress response genes, and downregulation of amino acid/protein synthesis, ribosome biogenesis, RNA metabolism and translation, and protein transport genes under treatment with Intestinal Like Medium (ILM) [43]. This difference can be partially explained by the loss of stress-responsive transcription factor (TF) binding sites (e.g., Msn2, Yap1, Hsf1, or Gcn4), coupled with the gain of new TF targets involved in carbon, nitrogen, and energy metabolism (e.g., Usv1, Adr1, Gzf3, or Hap1) in *Sb* promoters [43]. Metabolomics analysis of *Sb* under acid stress revealed upregulation of intracellular alanine, D-(+)-galactose, D-fructose-6-phosphate, glutathione (GSH), glycerol, and erythritol [50]. These findings suggest that the metabolic pathways, including glycolysis, fructose and mannose metabolism, glycerolipid metabolism, and tricarboxylic acid (TCA) cycle, play an important role in *Sb*'s acetate resistance. Therefore, these pathways represent potential targets for genetic engineering to enhance its acid tolerance.

The high production of antimicrobial acetate by *Sb* during glucose fermentation under anaerobic conditions contributes to its probiotic properties. For instance, Sb.A and Sb.P strains can secrete high levels of acetate at 37°C to inhibit the growth of *Escherichia coli* MG1655 [13]. Moreover, the cell-free supernatant of the ENT3 strain with higher acetate production displayed larger inhibition zone towards gut pathobionts such as Enterobacter, *Klebsiella pneumoniae*, *Clostridium perfringens*, Citrobacter, *Escherichia*/*Shigella*, and *Bacteroides fragilis* [52]. In addition, the residence time of *Sb* in mouse gut is positively correlated with higher acetate production [52]. It is hypothesized that the acidification of epithelial invaginations in the brush border, caused by the secreted acetate, may confer a competitive advantage to *Sb* over the other gut microbiota.

Quantitative trait locus (QTL) analysis revealed that mutant alleles of *SDH1* (encoding a flavoprotein subunit of succinate dehydrogenase in TCA cycle) and *WHI2* (encoding a phosphatase activator required for general stress response with plasma membrane Psr1 phosphatase) are found in 24 strains of *Sb* but not in *Sc* S288C [13]. The H202Y and F317Y mutations of *SDH1* (*sdh1*$^{\mathrm{H202Y,\ F317Y}}$) disrupt the succinate dehydrogenase complex at 37°C (Fig. 2), which leads to the impairment of TCA cycle [13]. The effect of the S287* mutation of *WHI2* is equivalent to the deletion of *WHI2*, which prevents the uptake and consumption of extracellular acetate at 37°C, leading to the accumulation of secreted acetate [13]. High acetate production trait in *Sb* requires two copies of *sdh1*$^{\mathrm{H202Y,\ F317Y}}$ alleles. On the other hand, the copy number of the *whi2*$^{\mathrm{S287*}}$ allele dictates transient and moderate (one copy) or continuous and high (two copies) accumulation of acetate. The reversal of either mutant *sdh1* or *whi2* alleles in *Sb* reduces its acetate production, whereas the introduction of the mutant alleles into *Sc* S288C increases the acetate production by 50% [13].

Comparative transcriptomics analysis between *Sc* and *Sb* discovered the downregulation of TCA cycle genes (*CIT3*, *IDP3*, *LSC2*, *SDH2*, *SDH5*, *SHH3*, *SHH4*, and *MDH3*) in *Sb* [43]. As the thermodynamic driving force (flux) is intrinsically associated with enzymes abundance, the

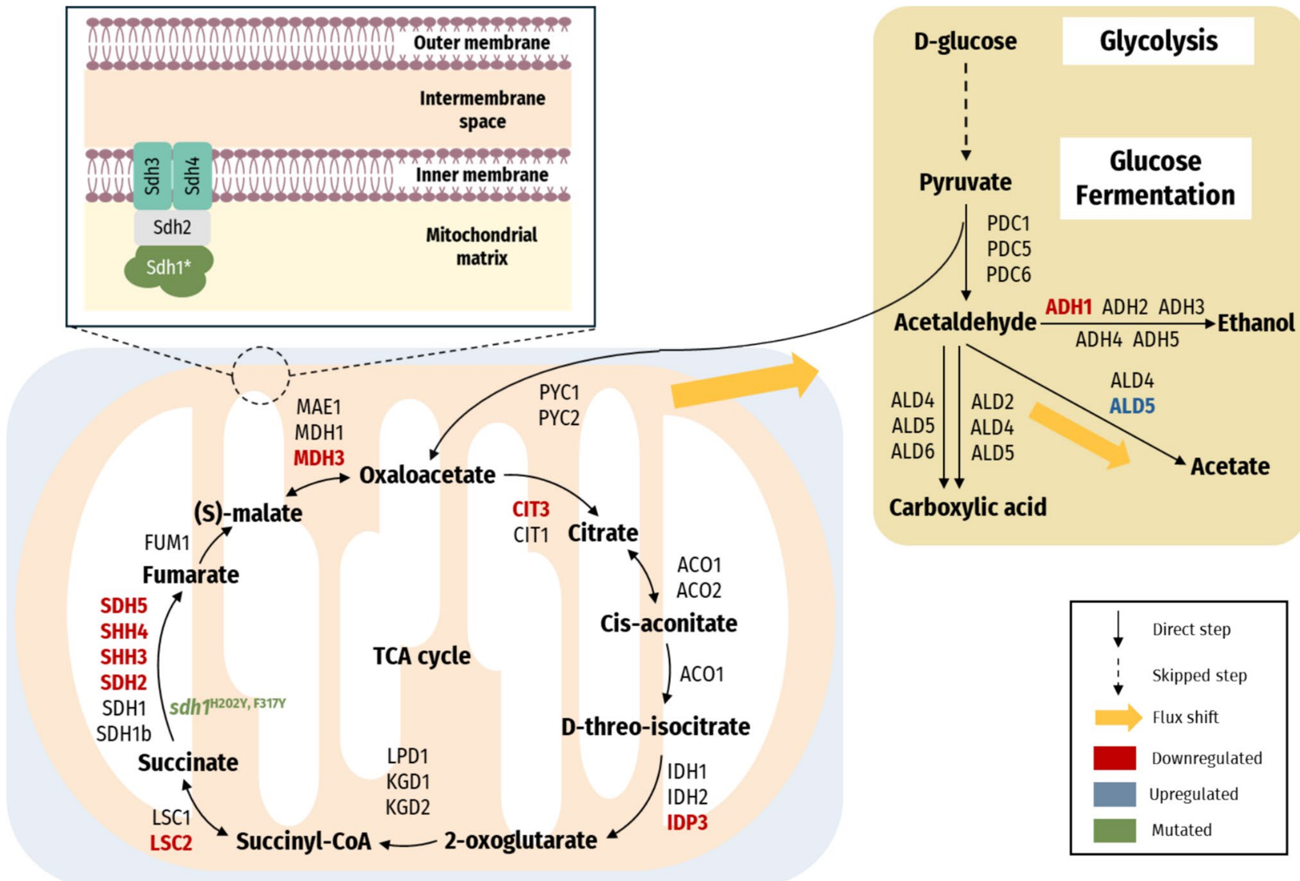

Fig. 2 A schematic representation of the TCA cycle, glycolysis, and the glucose fermentation pathway in *Sb*. Compared to *Sc*, *Sb* demonstrates transcriptional profiles that are indicative of enhanced acetate biosynthesis. For example, several genes in TCA cycle (*CIT3*, *IDP3*, *LSC2*, *SDH2*, *SDH5*, *SHH3*, *SHH4*, and *MDH3*) are downregulated in *Sb*, which results in a reduced flux through the TCA cycle. Besides, the mutant allele of *SDH1* (*sdh1*$^{\text{H202Y, F317Y}}$) also causes reduced flux through the TCA cycle. Additionally, the upregulation of *ALD5* enhances the conversion of acetaldehyde to acetate whereas the downregulation of *ADH1* further favors the production of acetate over ethanol.

downregulation of these TCA cycle genes restricts the flux through TCA cycle and creates a thermodynamic bottleneck. Due to the thermodynamic bottleneck, the total enzyme required to maintain flux through a metabolic pathway increases [53]. Consequently, the bottleneck imposes a metabolic burden on the cell, leading to flux shift toward more efficient ATP-generating pathways such as glucose fermentation [54]. In addition to the TCA cycle genes, the genes involved in glucose fermentation are found to be differentially expressed in *Sb* compared to *Sc*, which further enhance acetate production in *Sb* [43]. Specifically, the *ALD5* gene which encodes aldehyde dehydrogenase for the conversion of acetaldehyde to acetate is upregulated. Furthermore, the *ADH1* gene which encodes alcohol dehydrogenase that converts acetaldehyde to ethanol is downregulated. Figure 2 illustrates how these alterations enhance acetate production in *Sb*. Targeting genes involved in the TCA cycle and glucose fermentation thus represents a promising strategy for genetic engineering of *Sb* strains producing more acetate, as demonstrated by de Carvalho et al. [52].

Mechanisms of Probiotic Action

The three proven probiotic mechanisms of *Sb* are the luminal actions, mucosal actions, and trophic effects, as illustrated in Fig. 3.

Luminal and Mucosal Actions

Luminal actions by *Sb* include antimicrobial activity, prevention of bacterial binding and invasion of gut cells, and maintenance of healthy gut microbiota. Adherence of enterohemorrhagic *E. coli* (EHEC), *Salmonella Typhimurium*, *S. Typhimurium* DT104 mutant, and *S. Typhi* onto the surface of *Sb* cell wall was reported [55–57]. Although the larger size of *Sb* has been proposed to physically prevent pathogen

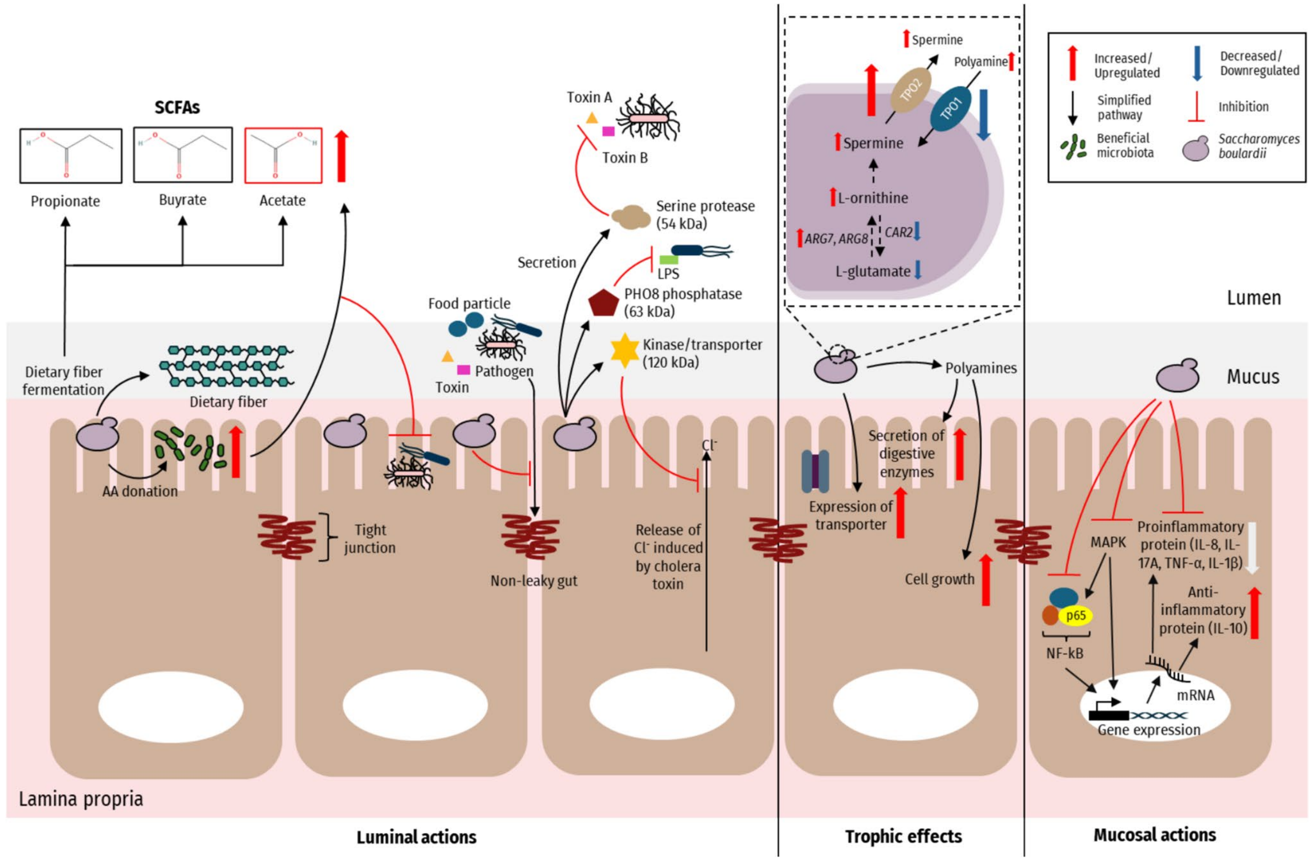

Fig. 3 Mechanisms of probiotic yeast *Sb* in maintaining gut health. Luminal action: *Sb* enhances concentrations of gut SCFAs directly via fermentation of dietary fiber and indirectly via synbiotic interactions with bacterial probiotics. One of the SCFAs produced is acetate, which has antimicrobial properties that help inhibit pathogen growth. *Sb* also secretes serine protease, PHO8 phosphatase, and kinase/transporter. The serine protease cleaves *C. difficile* toxins A and B, while the PHO8 phosphatase desphosphorylates the *E. coli* endotoxin lipopolysaccharide (LPS). The secreted kinase prevents chloride secretion in diarrhea caused by *Vibrio cholerae*. Trophic effect: *Sb* secretes polyamines such as spermine and spermidine that have signaling effects towards gut epithelial cells to enhance the synthesis of digestive enzymes and transporters. Mucosal action: *Sb* exerts anti-inflammatory effects through interference of NF-κB and MAPK pathways.

attachment onto the epithelial cell surface, some evidences do not support this hypothesis because *Sb* is unable to adhere to Caco-2 cells [8] nor reduce the adhesion of EHEC to T84 cells *in vitro* [58]. Besides, *Sb* is known to secrete proteins that neutralize pathogen toxins. For instance, a *Sb*-secreted serine protease is capable of cleaving *C. difficile* toxins A and B [46]. Furthermore, PHO8 phosphatase secreted by *Sb* can dephosphorylate endotoxin such as LPS from *E. coli* [48]. Moreover, a *Sb*-secreted 120 kDa kinase/transporter prevents cAMP- and Ca^{2+}-induced chloride secretion in T84 human carcinoma cells, which is associated with diarrhea caused by *V. cholerae* [59].

Apart from secreting antimicrobial proteins, *Sb* also ferments fiber to increase the intestinal levels of short-chain fatty acids (SCFAs). SCFAs are the carboxylic acids with two to six carbon atoms such as acetate (C2), propionate (C3), butyrate (C4), valerate (C5), and caproate (C6), which play crucial roles in local intestinal, psychological, and systemic health [60, 61]. As organic acids, SCFAs acidify and lower the intestinal pH to suppress pathogens proliferation. Besides, SCFAs can permeate into pathogen and dissociate into acid ($RCOO^-$) and hydrogen (H^+) ions. These ions can weaken the pathogen LPS outer membrane, increase intracellular osmotic pressure, and impose a metabolic burden to the pathogen for maintaining homeostasis [62]. Accumulating experimental evidence demonstrated that SCFAs can protect the integrity of tight junction and enhance intestinal barrier [62, 63]. For instance, butyrate promotes the transcription of tight junction protein Claudin-1, thereby enhancing epithelial barrier integrity. This was demonstrated *in vivo* using mice-derived Cdx2-intestinal epithelial cells, which are differentiated derivatives of undifferentiated IEC-6 cells stably transfected with the intestine-specific transcription factor Cdx2. These Cdx2-expressing cells form well-defined tight junctions and thus serve as a robust model for investigating tight junction dynamics and small intestinal barrier function [64]. The administration of butyrate also increases Mucin 2 (MUC2) formation which enhances the thickness of mucus layer that protects the mucosa [65] and induces the release of antimicrobial peptide CAP-18 [66].

On the other hand, propionate supplementation improves the intestinal barrier function in dextran sodium sulfate (DSS)–induced colitis mice model by inhibiting the decrease in expression of tight junction proteins such as zonula occludens-1, occludin, and E-cadherin in colonic tissue [67]. Aside from enhancing tight junction, propionate supplement also reduces chronic inflammation by reducing the expression of the pro-inflammatory cytokines (IL-1β, IL-6, and TNF-α) [67]. Propionate also reduces oxidative stress in colitis mice by reducing the colon level of oxidative stress promoting enzyme myeloperoxidase and increasing the colon and serum level of antioxidant enzymes superoxide dismutase and catalase [67]. Furthermore, propionate also inhibits the phosphorylation of STAT3 transcription factor, thus reducing inflammation and oxidative stress via the STAT3 signaling pathway [67]. Moreover, high acetate concentration demonstrates protective effects on the intestinal barrier by downregulating pro-inflammatory cytokines (IL-8 and TNF-α) and upregulating MUC2 of the ulcerative colitis patient-derived epithelial cells [68].

Other than fiber fermentation, *Sb* also indirectly influences the SCFA levels in the small intestine through symbiotic dynamics with the probiotic bacterial community (Fig. 3). For instance, strain ENT3 with extra high acetate production can preserve the gut microbiota richness in the inflamed mice gut [69]. Genome-scale metabolomic modeling of *Sb* with different probiotic bacteria by Hedin et al. [70] demonstrated that *Sb* reduces the overlap of metabolic resources and improves the metabolic interaction potential of the probiotics communities. Additionally, *Sb* donates amino acids to certain probiotic strains such as *Lactobacillus brevis* and *Bifidobacterium longum* subsp. *infantis*. As a result, the total bacteria number and acetate concentration increase after the introduction of *Sb*. In the mice model, the administration of *Sb* increases the abundance of beneficial bacteria (Akkermansiaceae and Bifidobacteriaceae) that produce acetate and butyrate [71]. Intriguingly, *Lacticaseibacillus rhamnosus*, commonly used alongside *Sb* in commercial probiotic formulations, competes with *Sb* according to *in silico* prediction and experimental validation [70]. This finding underscores the importance of comprehensive research on strain interactions to optimize the effectiveness of combined probiotic products.

Mucosal actions refer to the direct anti-inflammatory activities on the intestinal cells (Fig. 3). *Sb* exerts anti-inflammatory effects via interference of NF-κB and MAPK pathways [58]. A small heat-stable anti-inflammatory factor (< 1 kDa) of *Sb* is reported to be capable of blocking NF-κB activation and expression of IL-8 [72]. New evidence suggests that *Sb*'s anti-inflammatory effect may stem from its ability to produce acetate in high concentration (100 mM), which decreases the expression of pro-inflammatory cytokines such as IL-8 and tumor necrosis factor-α (TNF-α) in human epithelial cell cultures derived from ulcerative colitis patients [68]. Evidently, high acetate-producing *Sb* strains such as Sb.P and ENT3 can mitigate disease activity in the colitis mice model by downregulating the expression of pro-inflammatory cytokines (IL-1β, IL-2, IL-4, and TNF-α) and upregulating the anti-inflammatory cytokines (IL-10 and KC/GRO) to exert anti-inflammatory effects [69]. Meanwhile, the non-acetate-producing strain SDH1 exacerbates the gut inflammation [69]. In the constipated mice model, the administration of *Sb* decreases the expression of the pro-inflammatory cytokine IL-17 A, inducible nitric oxide synthase (iNOS), and p65 (a subunit that forms heterodimer with p50 to activate NF-κB signaling) [71]. Concurrently, *Sb* upregulates the expression of the anti-inflammatory cytokine IL-10 [71].

Furthermore, the administration of *Sb* also reduces expression of pro-inflammatory cytokines (IL-8 and TNF-α) and increases expression of the anti-inflammatory cytokine IL-10 in patients with irritable bowel syndrome [73]. Kan et al. [74] reported that *Sb* reduces gluten-induced immunopathology in immunized NOD/DQ8 mice via the activation of aryl hydrocarbon receptor (AhR) pathway, which regulates intestinal homeostasis and inflammation [75]. This mechanism involves the elevated expression of proteolytic genes and lowering the expression of tryptophan 2,3-dioxygenase due to *Sb*, which leads to increased free tryptophan and increased production of indole for activation of the AhR pathway [74]. Given that celiac disease (CD) patients exhibit reduced AhR pathway activation, modulating the tryptophan-*Sb*-host metabolic pathway represents a promising therapeutic management for CD [74]. Beyond its anti-inflammatory effects in the GI tract, *Sb* exhibits systemic anti-inflammatory properties by reducing the circulating pro-inflammatory cytokines throughout the body. For example, a recent study demonstrated that *Sb* not only attenuated colonic inflammation, but also reduced skin inflammation in the ovalbumin-induced atopic dermatitis BALB/c mice model by downregulating NF-κB [76]. Lowering of NF-κB activation restores epithelial membrane function and enhances tight junction protein expression, which strengthens the barrier against luminal inflammatory molecules.

Trophic Effects and Gut-Brain Axis Modulation

The trophic effects of *Sb*, extensively reviewed by Moré, Vandenplas [77], involve the stimulation of disaccharidase biosynthesis (lactase, sucrase, and maltase) [78, 79], nutrient transporters (such as sodium/glucose cotransporter-1 [SGLT-1]) [80], and immunoglobulins (IgA and its secretory component) [81]. While the precise mechanism remains unclear, it is believed that these effects are triggered by the endoluminal release of polyamines originating from *Sb*. The high polyamine content (~ 670 nmol/100 mg) in lyophilized *Sb* yeast cell [78], which is released during intestinal catabolism, can be absorbed by the brush-border membrane [82, 83]. Notably, both viable or killed forms of *Sb* have been shown to elicit trophic effects [79].

These polyamines—diamine putrescine, triamine spermidine, and tetra-amine spermine—are low-molecular-weight aliphatic polycations involved in signal transduction via the MAPK pathway [84]. *Sb* can stimulate the signaling cascade through GRB2-SHC-CrkII-Ras-GAP-Raf-$ERK_{1/2}$ to transmit mitogenic signals that promote cell proliferation and differentiation [85]. The trophic effect of *Sb* resembles the action of dietary polyamine administration, which stimulates disaccharidase biosynthesis [78] and promotes cell differentiation for intestinal maturation in a dose-dependent manner [86].

While earlier studies suggested that *Sb* cannot secrete spermidine and spermine [78], a recent comparative transcriptomics study comparing *Sb* and *Sc* revealed the upregulation of genes involved in polyamine biosynthesis and export. Specifically, the arginine biosynthesis genes (*ARG7* and *ARG8*) and spermine exporter gene (*TPO2*) were upregulated [43]. In contrast, genes related to the ornithine catabolism (*CAR2*), polyamine transport (*TPO1*), and spermine uptake (*PTK1*) were downregulated [43]. These expression patterns suggest that *Sb* is capable of synthesizing and secreting spermine and spermidine, thereby enhancing its trophic effects (Fig. 3). Together, these luminal, mucosal, and trophic actions help maintain the integrity of tight junctions between adjacent epithelial cells, thus preventing leaky gut, which allows the passage of undigested food particles, toxins, and microbes into the bloodstream.

The gut-brain axis (GBA) refers to the bidirectional communication between the enteric nervous system (ENS) of the gut and the central nervous system (CNS) of the brain, mediated by the gut microbiota, immune system, and the hypothalamic–pituitary–adrenal (HPA) axis. The ENS, often termed the "second brain", is a complex neural network spanning from the esophagus to the anal canal and functions autonomously to regulate gastrointestinal processes including motility, peristalsis, secretion, and vascular tone [87, 88]. Gut microbial communities modulate ENS activity by metabolizing nutrients, producing vitamins, and interacting with mucosal cells to influence neuronal excitability through diverse signaling pathways [89]. *Sb* has shown therapeutic potential in maintaining ENS structure and function. For instance, *Sb* treatment restored gut motility by correcting delayed ileal transit and accelerated colonic emptying in a murine model infected with Herpes Simplex Virus type 1 (HSV-1). Moreover, *Sb* preserved ENS integrity by attenuating peripherin damage, thereby countering HSV-1-induced neuroplastic alterations [90].

Imbalance in the gut microbiome, also known as gut dysbiosis, has been linked to cognitive-behavioral dysfunction in both human and animal studies [91]. The role of probiotics, particularly *Sb*, in protecting brain health has gained increasing attention in recent years. In a mouse model of antibiotic-induced gut dysbiosis, treatment with *Sb* effectively restored the gut microbiome, which had been disrupted by antibiotics. Specifically, *Sb* treatment restored the abundance of beneficial gut microbiota, such as *Lactobacillus*, *Bifidobacterium*, *Firmicutes*, and *Clostridium* [92]. This restoration was associated with reduced neuroinflammation, as indicated by decreased levels of pro-inflammatory cytokines (IL-1β, TNF-α, and MCP-1). Consequently, treated mice exhibited less hippocampal neuronal damage and improved cognitive-behavioral performance [92].

Furthermore, *Sb* treatment also ameliorated gut barrier impairment and corrected abnormalities in the fungal microbiota (mycobiota) in a microglia-induced neuroinflammation mouse model of Alzheimer's disease (AD) [93]. Specifically, *Sb* reduced the abundance of Aspergillaceae, which is associated with neuroinflammation. This effect was linked to the inhibition of microglia-induced neuroinflammation, evidenced by reduced levels of pro-inflammatory markers (CD11b, IL-1β, and IL-6), and the attenuation of the Toll-like receptor (TLR) signaling pathway, as indicated by the reduced expression of MyD88, TLR2, and TLR4 in AD mouse model [93]. These findings suggest that *Sb* can modulate the GBA to attenuate cognitive impairment in AD.

Additionally, *Sb* administration has been shown to reduce neuroinflammation in a pentylenetetrazole-kindled seizure mouse model. *Sb* treatment resulted in decreased hippocampal pro-inflammatory cytokines (IL-1β and IL-6) and increased levels of the anti-inflammatory cytokine IL-10, further supporting its role in mitigating neuroinflammation and promoting brain health [94].

Hyperactivation of the HPA axis, which includes the hypothalamus, anterior pituitary, and adrenal cortex, is frequently associated with depression [95]. In a cerebral palsy mouse model, *Sb* administration successfully reduced depression-like behaviors by lowering serum levels of pro-inflammatory cytokines (IL-6 and TNF-α) and HPA axis activity, as indicated by decreased adrenocorticotropic hormone and corticosterone levels [96]. A recent small-scale clinical trial demonstrated that a probiotic cocktail containing *Sb* led to significant improvements in depression symptoms compared to a placebo group [97]. These promising results highlight the potential of *Sb* as an adjunctive therapy for mental health disorders, although larger clinical trials are necessary to confirm these findings.

Furthermore, previous research has shown that *L. farciminis* can prevent gut permeability, thereby preventing HPA axis hyperactivation in LPS-challenged mice models [98]. Similarly, extensive studies have confirmed that *Sb* preserves gut tight junction integrity, which helps prevent the passage of LPS across the blood–brain barrier. This action mitigates LPS-induced memory impairment [99], anxiety [100], and oxidative stress [101] in mouse models. In summary, these findings highlight the multifaceted benefits of *Sb* in restoring gut microbiota balance, modulating neuroinflammation, and regulating HPA axis activity, all of which contribute to enhanced brain health.

Conclusion and Future Perspectives

Sb, a distinct probiotic yeast strain of *Sc*, exhibits unique genomic differences, including chromosome XII tetrasomy, the loss of intact Ty transposable elements, and copy number variations in specific genes. These genetic variations lead to phenotypic divergences, such as sporulation deficiency, altered carbon metabolism, and thicker cell walls. These features contribute to *Sb*'s enhanced thermotolerance and acid resistance, enabling it to thrive in the harsh gastrointestinal environment and function effectively as a probiotic. *Sb*'s ability to produce SCFAs, which possess antimicrobial properties, further supports its role in maintaining gut health. *Sb* has been demonstrated to neutralize luminal toxins, promote gut microbiome homeostasis, and modulate mucosal immune responses through the NF-κB and MAPK signaling pathways. Additionally, the trophic effects driven by polyamine secretion contribute to gut epithelial health, promoting the synthesis of digestive enzymes and nutrient transporters. Remarkably, current research also highlights *Sb*'s potential in supporting brain health by modulating the gut-brain axis. This emerging field underscores *Sb*'s role in neuroinflammation modulation, which has significant implications for cognitive function and mental health.

Advancements in synthetic and systems biology of *Saccharomyces* yeast have facilitated the bioengineering of *Saccharomyces* species [111], unlocking new possibilities for engineering *Sb* strains for the development of live biotherapeutics and advanced microbiome therapies [27, 102–109]. By integrating the natural probiotic properties of *Sb* with synthetic biology tools, *Sb* emerges as a promising multifunctional therapeutic agent for treating gastrointestinal disorders and addressing broader health challenges. In parallel, there is a growing trend in the application of *Sb* in the food processing industry, particularly for food development, probiotic fortification, and allergen detoxification [106–109, 112]. However, further research is required to address key concerns such as strain stability, biosafety, genetic containment, and clinical efficacy to ensure the safe and effective implementation of engineered *Sb* strains in therapeutic and industrial applications.

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Author Contribution H-H.G contributed to project administration, supervision, and funding acquisition. T-Y.T conceptualized, performed literature search and data compilation, and drafted the original draft. W-J.L drafted, reviewed, and critically revised the work. All authors reviewed the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval It is a review, and no ethical approval is required.

Animal Ethical Approval Not applicable.

Competing interests The authors declare no competing interests.

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